Journal of Applied Physics ARTICLE pubs.aip.org/aip/jap

Impact of mechanical scratches on the structural, thermal, and electronic properties of monocrystalline silicon

Cite as: J. Appl. Phys. **140**, 045102 (2026); doi: 10.1063/5.0324667
Submitted: 26 January 2026 · Accepted: 18 June 2026 ·
Published Online: 23 July 2026

View Online Export Citation CrossMark

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ABSTRACT

The influence of intrinsic and extrinsic defects on the electronic, thermal, and structural properties of monocrystalline silicon was investigated using photothermal radiometry (PTR), photoacoustic (PA) methods, x-ray diffraction (XRD), and electrical characterization. A p-type silicon wafer ($\rho = 0.005\ \Omega$ cm) was analyzed as a complete wafer, and a 1 cm^2 sample was cut from the wafer center. The 1 cm^2 silicon sample was examined before and after introducing one and two orthogonal central scratches. Mechanical defects were characterized by atomic force microscopy (AFM) to determine scratch width and depth. AFM measurements identify localized surface buckling and topographic deformation. In damaged regions, PTR and PA mapping revealed a decrease in thermal diffusivity from 0.97 to 0.15 $cm^2 s^{-1}$. XRD analysis showed a slight shift of the (400) diffraction peak from 69.124° to 69.146° and an increase in full width at half maximum from 0.049° to 0.060°, indicating lattice distortion. Structure refinement of diffraction patterns showed that the lattice strain increased from 0.103% before scratching to 0.295% after scratching. Electrical resistivity measurements using the van der Pauw method indicated increased disorder, reducing measurement reliability in cut samples due to structural non-homogeneity and disruption of the electron mean free path. The results establish a correlation between extrinsically induced mechanical lattice disorder and intrinsic defects due to carrier distribution, which causes the degradation of thermal and electronic transport, and highlight the sensitivity of PTR and PA techniques for the non-destructive detection of surface damage in silicon wafers.

I. INTRODUCTION

The fabrication of integrated devices and high-performance optoelectronic systems has relied extensively on silicon (Si) due to its availability, excellent electronic properties, thermal and chemical stability, and mature processing technology.[1,2] The structural properties and surface quality of Si wafers have consistently been shown to determine their functional performance. These properties are governed by intrinsic defects such as boron–oxygen complexes, variations in the carrier concentration and extrinsic defects (such as dislocations), which critically influence thermal transport, carrier lifetime, and optoelectronic response.[3]

In addition to intrinsic parameters, extrinsic defects introduced during wafer manufacturing and handling have been identified as equally important. Mechanical damage such as micro- and submicrometer-scratches, subsurface cracks, and cleaving defects from cutting, polishing, or sawing processes has been reported to act as scattering centers for phonons and recombination sites for carriers, thereby reducing minority-carrier lifetimes and altering phonon propagation.[4,5] Diamond wire sawing, in particular, has been associated with roughness and subsurface damage that strongly correlate with electrical performance degradation.[6]

In addition, it is important to consider defects arising from electronic circuits and fabrication processes, which inevitably introduce deviations from the ideal electronic structure of silicon. Although silicon is theoretically expected to exhibit a photoresponse limited to wavelengths below approximately 1100 nm, experimental evidence of out-of-band responses under near-infrared irradiation (e.g., 1319 nm) indicates that real devices do not behave as perfect intrinsic materials. Lü and Rongzhu[7] attribute this behavior to intrinsic point defects, such as vacancies and self-interstitials, which introduce localized energy levels within the bandgap and modify the optical absorption and quantum efficiency. First-principles calculations show that specific defect configurations, particularly tetrahedral interstitials, can strongly enhance infrared absorption, leading to bandgap narrowing or collapse, high absorption coefficients, and reduced saturation thresholds. While these results provide a compelling microscopic explanation for anomalous photoresponses, the analysis remains largely theoretical and relies on idealized defect models and concentrations that may not fully represent defects generated during wafer processing, circuit fabrication, or mechanical handling. Consequently, strong quantitative predictions would benefit from systematic experimental validation and correlation with realistic defect densities. This highlights the need for integrated experimental and theoretical approaches to assess how fabrication-induced defects impact the optoelectronic performance of silicon in high-precision detection and electronic applications.

Scratches and mechanical irregularities, whether intentional or unintentional, are critical limitations for advanced optoelectronic applications, as both macroscopic and localized wafer responses are affected. Device size has also become an increasingly important parameter. Reductions in device dimensions, or confinement in micro- and nanostructures, have been shown to modify thermal diffusivity, carrier recombination, and phonon-scattering mechanisms.[2,4] However, it is very important to consider changes in the wafer due to mechanical defects beginning with a complete characterization of the wafer.

Even under the most controlled conditions used for electronic device fabrication, the material remains far from behaving as an ideal wafer in terms of its structural and optothermal response. Before reaching the laboratory, the material undergoes several industrial processing steps, including ingot growth with a preferential crystallographic orientation, wafer slicing, polishing, and intrinsic doping. These processes inevitably introduce imperfections into the crystal structure, such as local lattice distortions, extended crystalline defects, and a non-uniform distribution of dopants. For p-type silicon wafers, boron incorporation is not ideally homogeneous, so even before any experimental manipulation, local variations in thermal and electronic response are already present.[8,9]

The non-destructive characterization techniques used in this study allow for the identification of structural imperfections and optoelectronic variations in silicon wafers. Photothermal and photoacoustic (PA) methods, such as photothermal radiometry (PTR)[10,11] and photoacoustic spectroscopy (PAS),[12,13] are highly sensitive to local changes in thermal transport and optical absorption, which are directly related to crystalline defects, surface damage, and recombination processes. These methods do not require electrical continuity or material isotropy, making them particularly suitable for mechanically scratched samples, where defects produce a highly localized and anisotropic response in thermal and electronic transport.

In this context, it is noteworthy that Hall effect measurements using the van der Pauw method do not yield reliable or reproducible values, indicating that the introduction of scratches creates electrical anisotropies and discontinuities that violate the assumptions of the model.[14,15]

Under these conditions, photothermal and photoacoustic techniques provide reliable information on crystalline quality and the local redistribution of absorbed energy. When combined with structural and spectroscopic techniques such as x-ray diffraction (XRD) and Fourier-transform infrared (FTIR) spectroscopy, this integrated approach enables the identification of imperfections in the crystalline structure and non-uniform dopant distributions, even when macroscopic electrical transport cannot be adequately described by idealized models.[3,16]

Photothermal and optoelectronic measurements have also been established as cost-effective, real-time methods for water quality control, linking defect characterization with industrial production requirements.[17] The voltage amplitude detected in the photoacoustic setup is related to the absorbed optical energy and subsequent heat diffusion, thus indirectly indicating carrier recombination and thermal transport.[16,17]

Photothermal and optoelectronic techniques exhibit high sensitivity, as the measured signal is generated by the absorption of incident radiation and its subsequent conversion and diffusion as heat within the material. Local alterations in the crystalline structure, such as intrinsic defects or mechanically induced scratches, modify optical absorption, non-radiative recombination processes, and thermal transport, leading to detectable changes in the amplitude of the photoacoustic signal. This direct dependence on fundamental energy-conversion mechanisms accounts for the effectiveness of these techniques in identifying highly localized structural variations.[18]

The objective of this work is to study the impact of intrinsic and extrinsic defects on the structural, thermal, and electrical properties of monocrystalline silicon wafers. Intrinsic defects result from carrier distribution within the silicon wafer, while extrinsic defects are caused by scratches. X-ray diffraction was used to identify changes in the crystal lattice induced by these defects by analyzing peak position shifts, line broadening, and localized damage in a small area ($1\ cm^2$), as well as by detecting new crystalline orientation produced by cutting. Photothermal radiometry and photoacoustic techniques were used to spatially resolve variations in thermal diffusivity and photothermal response associated with surface scratches, enabling the detection of local anisotropies in heat transport. The electrical resistivity of silicon samples was measured using the van der Pauw method to assess the influence of mechanical damage on charge transport and sample homogeneity. By correlating these complementary characterization techniques, this study establishes a direct relationship between structural damage, degradation of thermal transport, and limitations in electrical performance, providing a non-destructive framework for evaluating silicon wafer quality during early fabrication stages.

II. MATERIALS AND METHODS

A. Samples preparation

A p-type monocrystalline silicon {100} wafer was used, obtained from Polishing Corporation of America (PCA), and grown by the Czochralski (CZ) technique. The original wafer had a radius of 10 cm and a nominal resistivity of less than 0.005 Ω cm, as specified by the manufacturer.

A small square sample with an area of 1 cm^2 was cut from the central region of the complete wafer using a diamond tip and was named silicon-01 (Fig. 1). The sample had a thickness of 530 ± 10 μm and a single-side polished surface, while the opposite face remained unpolished to aid in optical and electrical orientation during measurement. The sample was analyzed in three stages, before and after mechanical scratching, as described in Fig. 1. Figure 1 shows the schematic sequence of sample conditions: from left to right silicon-01 corresponds to the sample without scratches [Fig. 1(a)]; silicon-02 corresponds to the sample after introducing a single vertical scratch at its center [Fig. 1(b)]; and silicon-03 is the sample with a second orthogonal scratch forming a cross-like pattern [Fig. 1(c)]. The marked points in each image indicate the reference positions where photoacoustic and photothermal spectroscopies were performed to measure local variations in thermal diffusivity and signal amplitude.

B. Mechanical scratch characterization

The scratch profile was measured using an atomic force microscope (AFM) (Nano surf FlexAFM, model C3000 controller). Measurements were performed in the tapping mode with a silicon tip at a setpoint of 50%. Surface scans were covered areas of $100 \times 100\,\mu$m, with 256 scan lines recorded at 0.6 s per line.

C. X-ray diffraction analysis

Structural analysis was performed using a Rigaku Ultima IV x-ray diffractometer. Diffraction patterns were obtained at an accelerating voltage of 35 kV and a current of 15 mA, using Cu Kα radiation with a wavelength of λ = 0.1540 nm. Data were collected over a 2θ range of 20°–80° with a step size of 0.02°.[16] The structural study focused on silicon wafers analyzed before and after the introduction of intentional mechanical damage. Additionally, x-ray diffraction measurements were performed on powdered silicon to identify and index all crystalline planes associated with the silicon crystal structure. MDI JADE 6 software was used to determine the stress on the structure due to the mechanical damage. Additionally, the PDF references corresponding to the crystalline phases identified in the samples were corroborated using Match! Software and PDF 4+ database, which were used for peak analysis to determine the structural phases present.

D. Photothermal and photoacoustic theory

Photothermal (PT) and photoacoustic (PA) spectroscopies are non-destructive techniques widely used to investigate the thermal, electronic, and structural properties of metals and semiconductors because of their high sensitivity in detecting defects and local variations in material properties.[11] Both methods are based on the absorption of modulated optical radiation at the sample surface, which generates a periodic temperature field. The resulting thermal diffusion wave propagates through the material, carrying information about its thermophysical parameters.[16,19]

The theory of the photoacoustic effect in solids, based on the Rosencwaig and Gersho model, was used to determine thermal diffusivity mapping. The photoacoustic signal is governed by the thermal regime of the material, which depends on the relationship between the thermal diffusion length (μ), the optical absorption coefficient (β), and the sample thickness (l). In the thermally thin regime, Eq. (1), the sample thickness is smaller than the thermal diffusion length ($l < \mu$); conversely, when ($l > \mu$), the material is in the thermally thick regime, Eq. (2). Additionally, relationship between l and optical absorption length (μ_β) determines whether the sample behaves as optically transparent or optically opaque. Detailed information about the experimental procedure is provided in the corresponding section,[16,20]

$$V_{\text{thin}} = \frac{A}{\sqrt{(1+2\pi f \tau_M)^2}} \cdot \frac{1}{\sqrt{f}}, \tag{1}$$

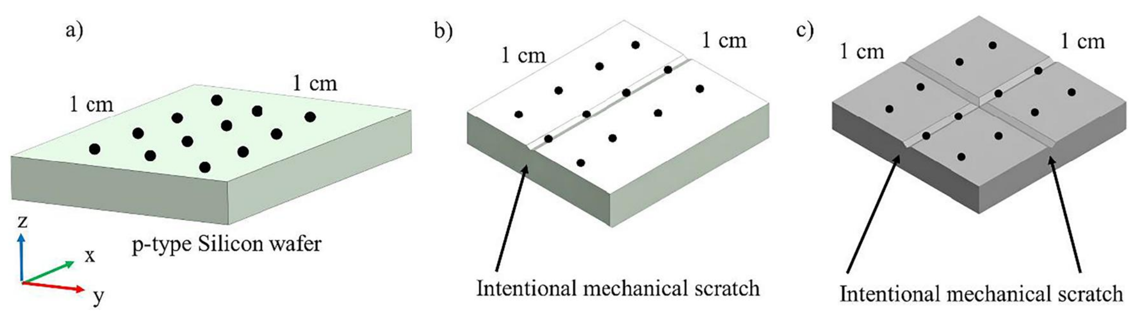

FIG. 1. Schematic representation of (a) silicon-01, (b) silicon-02, and (c) silicon-03.

$$V_{\text{thick}} = \frac{A}{\sqrt{1+(2\pi f \tau_M)}} \cdot e^{-a\sqrt{f}}, \tag{2}$$

where f is the modulated frequency of the laser, τ_M is the time response of the microphone, and A is a parameter that includes the pressure, the thermal properties of the air in the acoustic chamber, the light intensity, and the geometry of the photoacoustic cell. The thermal diffusion length is expressed as

$$\mu = \sqrt{\frac{2\alpha}{\omega}}, \tag{3}$$

where a is a constant, ω represents the angular modulation frequency ($2\pi f$), and α denotes the thermal diffusivity.

In semiconductors such as silicon, heat transport is mediated by phonons and charge carriers, and the interaction between them determines the propagation of the thermal wave. Variations in diffusivity or phase shift of the PTR and PA signals are indicative of structural defects, electronic recombination centers, or surface damage, allowing correlations between optical and thermal responses and crystalline quality.[21,22]

The frequency-domain photoacoustic (FDPA) approach has proven highly effective in characterizing monocrystalline silicon wafers, enabling the detection of local variations in thermal diffusivity, carrier lifetime, and defect distribution. When used alongside complementary techniques such as the Hall effect and x-ray diffraction, FDPA offers a comprehensive evaluation of the thermal and electronic behavior of silicon before and after surface modifications.[6,16]

Therefore, experimental measurements were conducted to determine the spatial distribution of thermal diffusivity and the photoacoustic signal amplitude across the silicon samples. The methodologies implemented were designed to operate within the appropriate thermal regimes and to capture local variations induced by surface modifications.

Equation (1) is used to determine the time response of the microphone. Then, the value τ_M is substituted into Eq. (2). This equation is fitted in order to the measured sample to obtain a constant. This constant is correlated with the thermal diffusivity by

$$\alpha = \frac{\pi l^2}{a^2}, \tag{4}$$

where α is the thermal diffusivity and l is the thickness of the sample.

To determine PA thermal diffusivity values, for each point measured, the thickness was measured. The thickness of the medium was $530 \pm 10\,\mu$m.[16]

E. Photothermal (PTR) methodology

Photothermal radiometry (PTR) was used to obtain the thermal images of silicon samples. The PTR technique is based on periodic optical excitation and the subsequent detection of modulated infrared emission associated with thermal relaxation, while background Planck radiation is minimized through phase-sensitive detection. The system operates in the mid-infrared spectral range (2–12 μm). Thermal information was obtained from two channels, PTR amplitude and phase, using a tightly focused laser beam with a spot size of approximately 40 μm. Optical excitation was provided by a high-power semiconductor laser ($\lambda = 880$ nm, $P = 300$ mW), which was collimated and focused onto the sample surface using a Gradium lens. A modulation frequency of 99 Hz was chosen to ensure operation in the thermally thick regime.[16] The sample was mounted on a motorized XY translation stage (Melles Griot) to enable the spatial mapping of the PTR amplitude and phase signals across the entire surface.

The modulated infrared emission generated by non-radiative relaxation processes was collected using two off-axis parabolic mirrors, collimated, and focused onto a liquid-nitrogen-cooled HgCdTe detector (Judson J15D12-M204). The detector output was amplified with a low-noise preamplifier and demodulated using a lock-in amplifier (SRS-850), which was interfaced with a computer for data acquisition and processing.[22]

F. Photoacoustic and photoacoustic images methodologies

An open photoacoustic (PA) cell was used to determine the thermal diffusivity of the samples. Optical excitation was provided by a high-power semiconductor laser ($\lambda = 880$ nm, P = 300 mW), which was collimated and focused onto the sample surface using a Gradium lens. To extract the thermal diffusivity, a frequency scan from 20 to 200 Hz was performed to probe the thermally thin and thermally thick regimes [Eqs. (1) and (2)]. An electret microphone was positioned at the rear of the sample to detect the generated acoustic waves. The microphone signal was passed through a low-pass filter, following the methodology reported by Martinez-Munoz and then processed with a lock-in amplifier (SRS-850) interfaced with a computer for data acquisition and analysis.[23]

The procedure described by Martinez-Munoz was used to determine the microphone time response and calculate the thermal diffusivity of the silicon samples.[16] Twelve points were measured on each sample before intentional scratching. At each position, both PTR and PA amplitudes (voltage output) were recorded. After introducing a scratch approximately 1 cm long at the wafer center with a diamond tip, measurements were repeated at the same locations. The voltage signals were correlated with local optical absorption and heat diffusion processes; variations in the amplitude and phase were interpreted as changes in local thermal diffusivity. The thermal diffusivity (α) was recalculated at each point using the fitted PA model and the known sample thickness. All experiments were conducted under controlled environmental conditions.

G. Electrical properties

The van der Pauw method was used to measure the resistivity before and after the intentional mechanical damage on the silicon samples. The methodology was according to the ASTM F76-86 norm, which consists of a system that determines current and voltage measures at different positions on the sample. The maximum range of current injections must be ± 10 mA, and the maximum range of voltage must be ± 500 mV

TABLE I. Voltage values determined by the Hall effect using four-point methodology, to determine the Hall coefficient.

$V_{24,13,B^+}$ (mV)	$V_{13,42,B^+}$ (mV)	$V_{24,13,B^-}$ (mV)	$V_{13,42,B^-}$ (mV)	$V_{42,13,B^+}$ (mV)	$V_{31,42,B^+}$ (mV)	$V_{42,13,B^-}$ (mV)	$V_{31,42,B^-}$ (mV)
−0.000 384 2	−0.001 237 8	0.002 649	0.002 072 2	0.001 207 8	0.000 364 2	0.00056	0.001 327 2

Instruments used to determine resistivity were a multimeter Keithley 2000, a current source Keithley 6220, and a picoammeter Keithley 6485, and a Keithley 7001 scanner to control the current injection and the voltage measurement on the four points in the sample.[6,16]

To determine the carrier mobility, it is necessary to calculate the Hall coefficients R_{HC} and R_{HD} using the following equations:

$$R_{HN} = \frac{2.5x10^7 \cdot l \cdot [(V_{31,42,B^+}) - (V_{13,42,B^+}) + (V_{13,42,B^-}) - (V_{31,42,B^-})]}{B \cdot I}, \quad (5)$$

$$R_{HP} = \frac{2.5 \times 10^7 \cdot l \cdot [(V_{42,13,B^+}) - (V_{24,13,B^+}) + (V_{24,13,B^-}) - (V_{42,13,B^-})]}{B \cdot I}, \quad (6)$$

where (R_{HN}) and (R_{HP}) represent the Hall coefficients determined for each measurement configuration, expressed in cm^3/C. l denotes the sample thickness (cm), and (B) is the magnetic field applied perpendicular to the sample surface, expressed in Tesla (T); in this study, a magnetic field of 0.56 T was used.[6,16]

Voltages were measured for different contact configurations under both positive (B^+) and negative (B^-) magnetic field polarities, according to the ASTM F76-86 norm. The factor (2.5×10^7) corresponds to the unit conversion required to express the Hall coefficient in cm^3/C, consistent with the measurement system used in this study. Table I presents the voltages obtained from the four-point configuration.

III. RESULTS AND DISCUSSION

A. Intrinsic defects on silicon wafer

Figure 2 shows the thermal diffusivity values at various locations across the silicon wafer, measured by the photoacoustic technique, including both the center and the periphery, to evaluate the intrinsic thermal transport properties before electronic device fabrication. The measured diffusivity values exhibit a clear spatial distribution, ranging from approximately 0.6 to 1.2 $cm^2 s^{-1}$. This variability is attributed to non-uniform dopant distribution within the wafer, as dopant atoms introduce local lattice perturbations that affect phonon transport without disrupting the long-range crystalline order. The red square in Fig. 2 represents the region selected for the analysis of thermal properties before and after the introduction of a mechanical defect. This area corresponds to the zone where the thermal diffusivity range is narrow. The measured values are in the range of 0.75 to 0.95 cm^2/s, providing a suitable reference region for evaluating changes in structural and thermal properties induced by extrinsic defects. As reported by Martinez-Munoz *et al.*, such dopant incorporation is inherently non-homogeneous across the wafer due to limitations in the manufacturing process. Therefore, the observed thermal diffusivity distribution, for all the wafers, reflects intrinsic material inhomogeneities that exist before device processing and must be considered when assessing the baseline thermal behavior of semiconductor substrates.[6]

B. Surface profilometry analysis of the scratch-induced deformation

Figure 3(a) shows the profilometry map obtained over a $100 \times 100\,\mu$m area at the center of silicon-02. The value

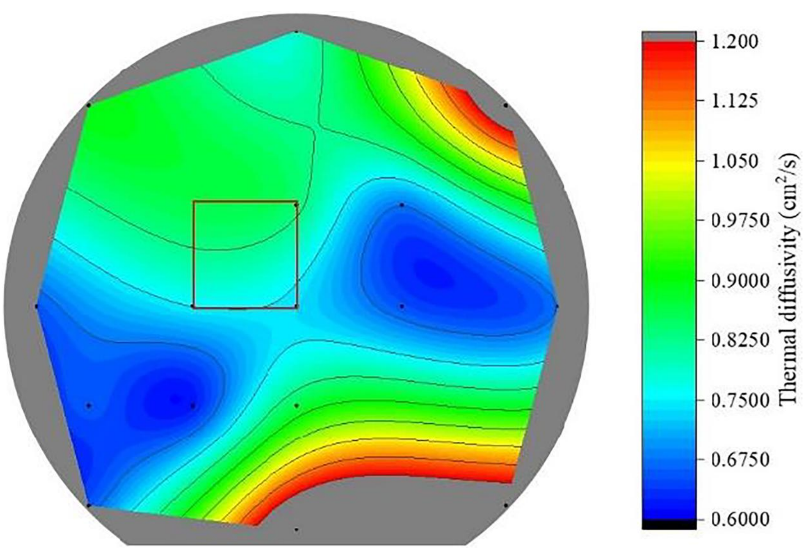

FIG. 2. Thermal diffusivity mapping of a p-Si wafer (0.005 Ω cm) obtained by photoacoustic technique. Red square represents 1 × 1 cm^2 space cut and analyzed.

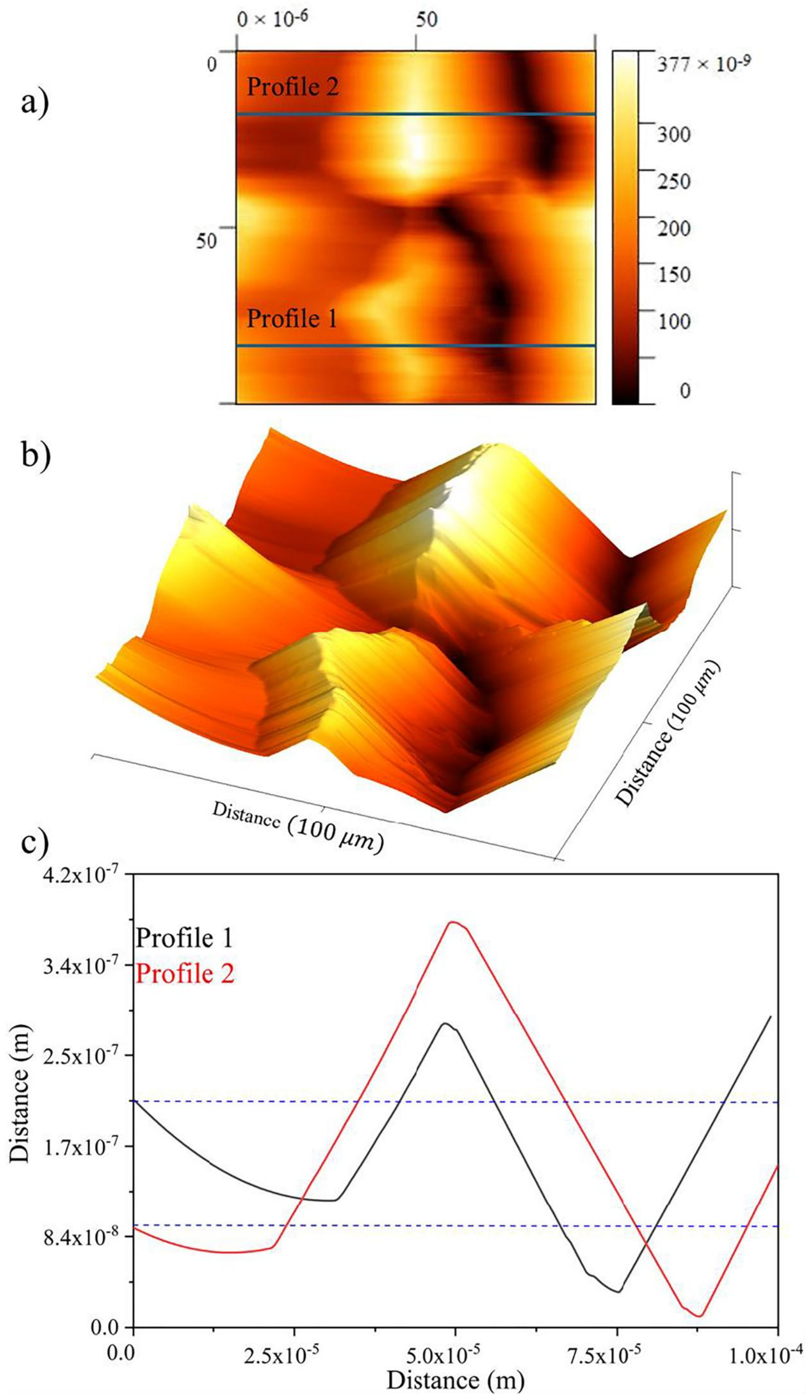

FIG. 3. Profilometry mapping over a 100 × 100 μm^2 area (a), surface relief mapping of the same region (b), and line profiles extracted from two selected zones (c).

distribution represents the surface height variation across the scratch. In this map, the lowest values correspond to the deepest regions of the scratch, while higher values correspond to surface deformation produced by the diamond tip, generating local height variations. Figure 3(b) shows the corresponding three-dimensional topographic representation. The scratch morphology is irregular. The scratch path does not follow a perfectly defined linear trajectory, indicating non-uniform mechanical interaction between the diamond tip and the silicon surface, which produces localized structural deformation. Figure 3(c) shows the relief profiles

TABLE II. Values determined from the scratch profiles, including height reduction and height variation measured along two-line profiles of the mechanical scratch.

	Y min (μm)	Y max (μm)	Silicon wafer baseline (μm)	Height reduction (μm)	Height variation (μm)
Profile 1	0.326	2.81	2.11	1.784	0.7
Profile 2	0.1017	3.755	0.921	0.8193	2.834

extracted from two regions marked in Fig. 3(a). In these profiles, the position at 0 μm on the x axis corresponds to the baseline surface of the wafer. As the lateral distance increases, a clear height variation is observed, characterized by an initial deformation region followed by a height reduction corresponding to the scratch valley. The two profiles exhibit different depth and width values, which can be attributed to variations in the applied force during the scratching process, resulting in non-uniform mechanical damage along the scratch path.

Table II summarizes the dimensional parameters extracted from the profiles shown in Fig. 3(c). The values (Y_{Max}) and (Y_{Min}) correspond to the maximum and minimum heights measured for each profile across the scratch. The silicon wafer baseline served as the reference height to evaluate the surface modifications caused by the mechanical defect; this baseline corresponds to the blue reference lines in Fig. 3(c). Using this reference, the height reduction and height variation were calculated from the differences between (Y_{Max}) and (Y_{Min}) relative to the wafer baseline. These parameters quantify the surface deformation induced by the diamond tip, describing both the valley depth below the baseline and the material displacement above the original silicon surface.

As can be observed, the induced mechanical scratch creates localized folds and surface distortions on the silicon wafer. Because the defect is not spatially consistent, it produces non-uniform structural strain within the lattice. These strain fields introduce lattice disorder, which enhances phonon scattering and reduces the phonon mean free path (MFP). Consequently, heat and carrier transport properties are affected, as both thermal diffusion and electronic mobility depend strongly on the integrity of the crystal lattice.

C. X-ray diffraction analysis

Figure 4 shows the x-ray diffraction patterns of silicon-01 [Fig. 4(a)] and silicon-02 [Fig. 4(b)]. In both patterns, the crystal phases were identified using the corresponding PDF cards,

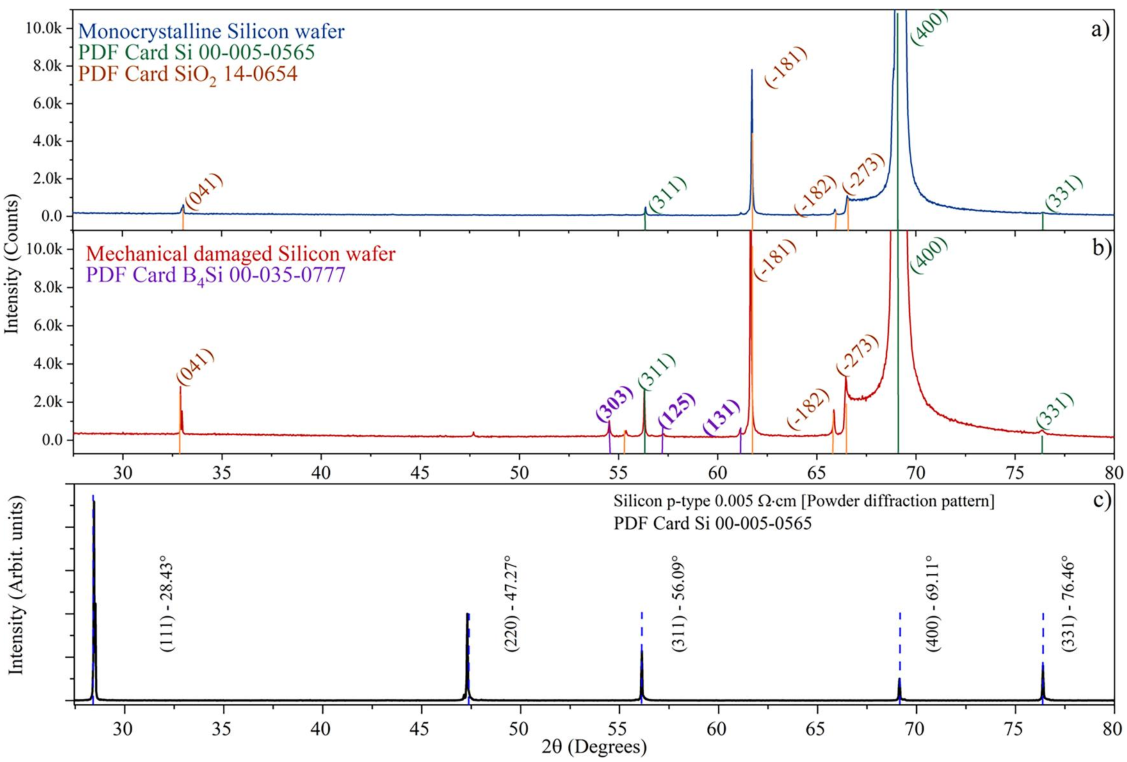

FIG. 4. X-ray diffraction patterns of (a) a monocrystalline silicon wafer and (b) a monocrystalline silicon wafer with mechanical damage. (c) corresponds to the powder diffraction pattern of crystalline silicon p-type.

identifying Si (PDF 00-005-0565) and SiO_2 (PDF 14-0654). The diffraction peaks for Si appear in both cases. However, after cutting samples with mechanical damage, some new diffraction peaks appear. This confirms that extrinsic cutting causes significant damage to the crystal structure of the sample. Additionally, in monocrystalline silicon wafer [Fig. 4(a)] only the Si and SiO_2 phases are detected, whereas in the sample with mechanical damage [Fig. 4(b)], an additional diffraction contribution associated with the B_4Si phase (PDF 00-035-0777) is observed. This suggests that additional mechanical damage promotes the appearance of new diffraction peaks that influence thermal and electrical properties. Figure 4(c) shows the powder diffraction pattern of p-type silicon (0.005 Ω cm). The diffraction pattern exhibits reflections corresponding to diamond-cubic silicon. As can be observed, all diffraction planes are identified in silicon-01 and silicon-02, establishing a correlation between the allowed planes associated with SiO_2 and B_4Si phases. The weak peaks adjacent to the Si (311), (004), and (331) reflections correspond to CuK_β signals rather than the high-index reflections currently assigned.

Figure 5 shows the (400) diffraction peak of silicon-01 and silicon-02. The peak profiles were analyzed to determine the FWHM. The (400) diffraction appears at 69.124° for the silicon-01 sample and exhibits and shifts slightly to 69.146° after additional mechanical damage (silicon-02).

A comparison of the (400) diffraction peak shows that both samples have similar peak intensities, indicating that overall crystallinity and preferred orientation remain essentially unchanged after mechanical damage. The small shift in the 2θ position can be interpreted using Bragg's law, which relates the peak position to interplanar spacing. Although no measurable variation in the lattice parameter is detected, the observed peak shift is attributed to residual structural stress induced by mechanical damage, consistent with previous reports by Martinez-Hernandez *et al.*, 2022.[6]

The (400) diffraction peak of the silicon-01 sample is located at 69.124°, while the silicon-02 sample shows a slight shift to 69.146°. According to Bragg's law, this change reflects a modification in the diffraction condition associated with variations in interplanar spacing. In addition to the peak displacement, a clear broadening is observed, with the full width at half maximum (FWHM) increasing from 0.04901° (silicon-01) to 0.0601° (silicon-02). This peak broadening indicates lattice distortion and defect formation caused by mechanical processes such as scratching or cutting. Together, the shift in the peak position and the increase in the FWHM demonstrate that intentional mechanical damage introduces lattice damage and disorder while preserving the overall monocrystalline nature of the silicon wafer.[6,16,19]

Structural strain was determined by refining the x-ray diffraction patterns using the corresponding PDF and CIF files for each crystalline phase identified in the diffractogram. For the monocrystalline silicon wafer (silicon-01) and the mechanically damaged silicon wafer (silicon-02), the calculated strain values were 0.103% and 0.295%, respectively. These values quantify the degree of lattice deformation caused by mechanical damage.

The increase in strain is directly associated with greater lattice distortion, which affects the phonon mean free path that governs heat transport and, consequently, influences electron mobility. The introduction of extrinsic defects reduces the phonon mean free path, thereby modifying the thermal and electronic transport properties of monocrystalline silicon.

D. Photothermal image and thermal diffusivity

Figures 6(a) and 6(b) show the photothermal radiometry (PTR) amplitude and phase signals measured at all points indicated in the schematic in Fig. 1(a) for a silicon-01 sample. The results are shown using a color map, ranging from low (blue) to high (red) values. As shown in Figs. 6(a) and 6(b), the distribution of the measured values is inhomogeneous, with a slight tendency toward lower values. These measurements serve as the reference baseline for evaluating variations in the thermal properties induced by intentional mechanical scratches. Figures 6(c) and 6(d) show the PTR amplitude and phase signals for the silicon-02 sample with a single vertical scratch, measured at the same twelve points previously analyzed in Fig. 1(b).

The AFM analysis shows that the scratch creates a surface relief with a characteristic lateral scale of approximately ∼100 μm,

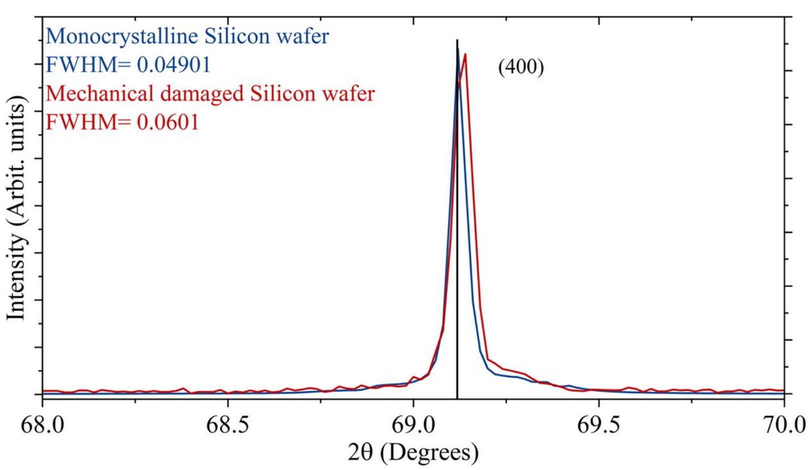

FIG. 5. Comparison of the (400) diffraction peak position for silicon-01 and silicon-02, a family of the (100) plane.

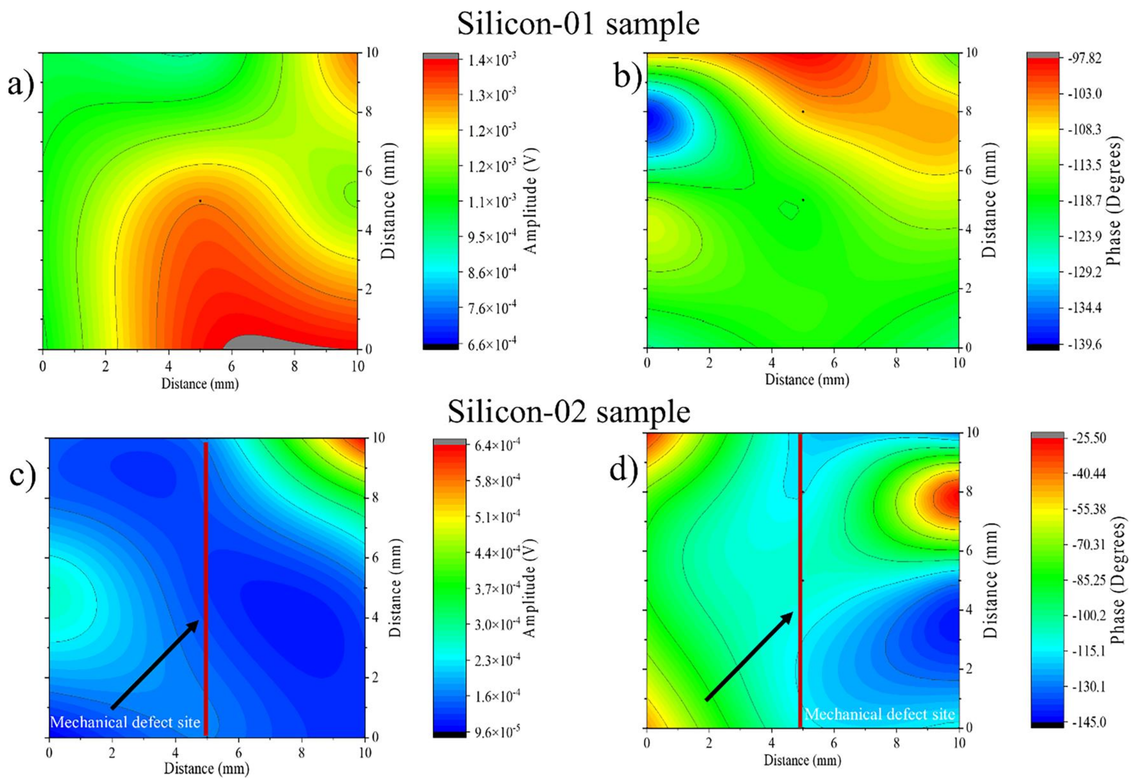

FIG. 6. Photothermal radiometry (PTR) amplitude and phase signals for (a) and (b) silicon-01 and (c and d) silicon-02.

consistent with the expected width of the mechanical damage caused by the diamond tip. The topographic profiles (Profile 1 and Profile 2) show pronounced surface deformation, with a local ridge or surface buckling structure forming on one side of the scratch.

This localized surface buckling is especially evident around the defect [Figs. 3(a) and 3(b)], where the AFM profiles indicate a significant height variation. Such deformation can modify the local thermal boundary conditions by changing the effective surface geometry and the heat diffusion pathways. As a result, the propagation of the thermal diffusion wave generated during PTR measurements may be delayed or distorted in this region. This behavior is consistent with the phase map observed in Fig. 6(d), which shows noticeable phase retardation near the scratch location.

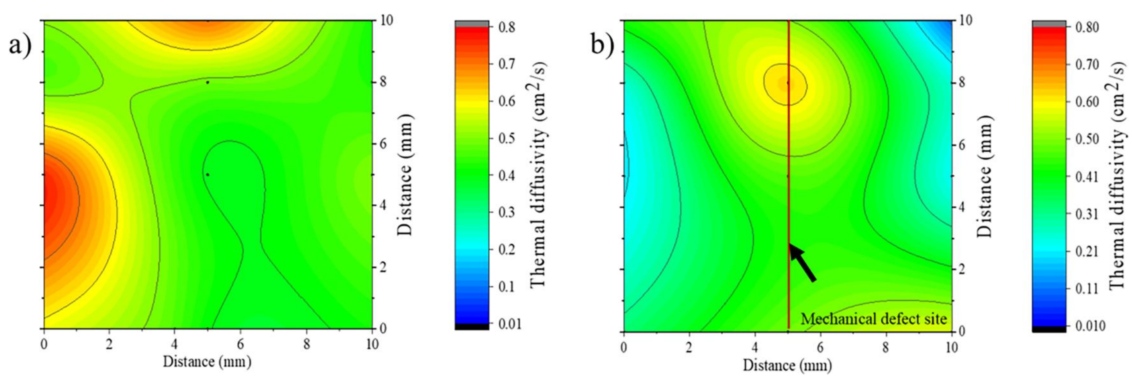

FIG. 7. Thermal diffusivity value distribution for (a) silicon-01 and (b) silicon-02.

Figure 7 shows the effect of mechanical damage on the thermal diffusivity on the silicon-01 sample. In the silicon-01 sample, shown in Fig. 7(a), the thermal diffusivity values are spatially non-uniform, reflecting spatial variations in thermal

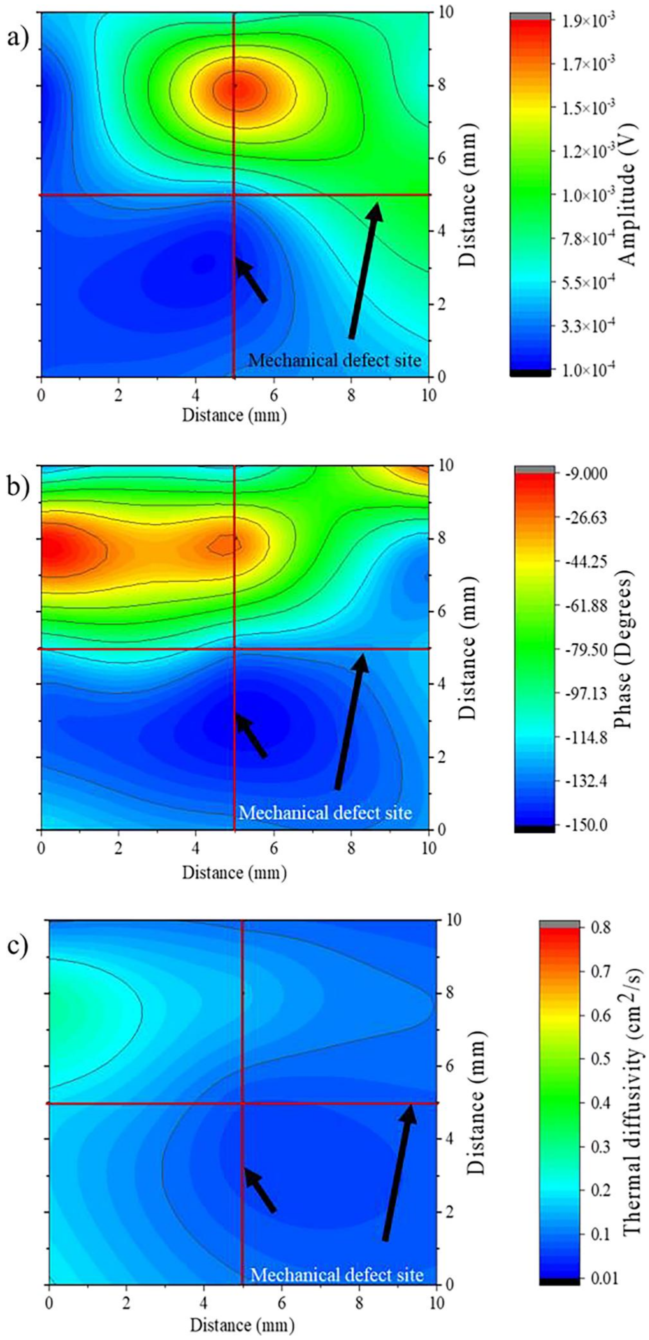

FIG. 8. (a) PTR amplitude signal, (b) PTR phase signal, and (c) PA thermal diffusivity values for silicon-03 sample.

transport properties. After the introduction of the orthogonal scratch (silicon-02), shown in Fig. 7(b), the thermal diffusivity distribution becomes clearly heterogeneous. A marked decrease in diffusivity is observed in the region associated with the mechanical defect, while the surrounding areas retain higher values. This contrast highlights the influence of the scratch on local heat transport, revealing how surface damage modifies the thermal response of the material.

Figure 8 shows the PTR amplitude and phase responses, as well as the thermal diffusivity, of the silicon-03 sample, which has two centered orthogonal scratches on the sample. The PTR amplitude map in Fig. 8(a) shows a pronounced perturbation in the central region of the wafer, where the scratches intersect, indicating a strong local modification of the thermoelastic response. In contrast, regions away from the damaged area exhibit lower and more uniform amplitude values, consistent with relatively undisturbed thermal transport. The corresponding phase and amplitude distributions in Figs. 8(a) and 8(b) show a clear spatial reorganization of the PTR signal, with a directional pattern following the orientation of the second mechanical scratch, indicating anisotropic heat diffusion induced by the damage. Additionally, the photoacoustic thermal diffusivity map in Fig. 8(c) shows a systematic decrease in thermal diffusivity across the damaged region, reflecting increased lattice disorder and enhanced phonon scattering caused by the orthogonal mechanical scratches, while the surrounding intact areas retain higher diffusivity values characteristic of crystalline silicon.

In the PTR amplitude maps, the silicon-01 sample appears mainly in intermediate tones, ranging from $10^{-4} - 10^{-3}$ V, with slightly higher values at the edges, as shown in Fig. 6(a). In silicon-02, the amplitude visibly decreases along the defect to about 10^{-4} V, as shown in Fig. 6(c). In the silicon-03 sample, a localized peak emerges at the intersection ($> \sim 1.8 \times 10^{-3}$ V), while the periphery drops below 3.2×10^{-4} V, as shown in Fig. 8(a).

For the phase signal, the values for the crystalline silicon sample range from $-97°$ to $-139°(\theta)$, exhibiting a generally spatially consistent distribution across the surface, except at the upper boundary, where the highest phase values are observed, as shown in Fig. 6(b). After the first scratch is introduced, the phase range broadens to between $-25°$ and $-135°$ (θ), and a distinct central region aligned with the scratch direction becomes evident, maintaining nearly constant phase values. This modification alters the surrounding phase distribution, suggesting that the central scratch suppresses regions of higher phase response, as shown in Fig. 6(d). For the silicon-03 sample, with two orthogonal central scratches, the phase values vary from $-9°$ to $-140°$ (θ), showing a clear orientation of the highest values that follows the direction of the second mechanical damage, as shown in Fig. 8(b).

In the thermal diffusivity maps, the silicon-01 sample, Fig. 7(a), is concentrated around $0.35 - 0.55$ cm^2/s, with edges reaching $\sim$0.7 cm^2/s. In the silicon-02 sample, Fig. 7(b), a slight reorganization appears at around $\sim$0.5 cm^2/s in the central region. In the silicon-03 sample, Fig. 8(c), the intersection region drops below 0.15 cm^2/s, showing a notable thermal decrease compared to the rest of the surface.

A more detailed analysis reveals that, taking an average silicon-01 value of $\sim$0.45 cm^2/s, the diffusivity in the damaged region with two scratches, silicon-02, decreases by about 67% $\left(\text{to less than } 0.15\ \frac{\text{cm}^2}{\text{s}}\right)$. Comparing the silicon-01 edge maxima (about 0.7 cm^2/s) with the silicon-03 final minimum value shows a reduction of ~79%. In PTR amplitude, a central reference value of $\sim 1.0 \times 10^{-3}$ V in the silicon-01 state decreases along the first scratch, silicon-02, to $\sim 1.7 \times 10^{-4}$ V (a reduction of $\sim 80\% - 85\%$), while at the intersection of two scratches, silicon-03, it exceeds 1.8×10^{-3} V (~80% higher than the reference), and the final periphery falls below 3.2×10^{-4} V ($\sim 65\% - 70\%$ lower). Furthermore, the measured values for the scratch-free silicon-01 ($0.35 - 0.55\ \text{cm}^2/\text{s}$) are below the typical range reported at 300 K for this material ($0.8 - 0.9\ \text{cm}^2/\text{s}$).[24]

The photothermal radiometry (PTR) and photoacoustic (PA) responses can be coherently explained by the structural modifications identified by x-ray diffraction. The progressive transition from spatially consistent to spatially non-uniform thermal and photoacoustic patterns with increasing mechanical damage reflects the gradual introduction of lattice imperfections and residual stress. In the silicon-01 sample, the absence of detectable lattice distortion results in uniform heat propagation and optical absorption. Introducing a scratch (silicon-02) causes a moderate decrease in thermal diffusivity in the central region, accompanied by the attenuation of the PA amplitude, indicating the partial disruption of phonon transport without significant thermal confinement. In contrast, two perpendicular scratches (silicon-03) cause a pronounced degradation of thermal transport, with diffusivity values dropping below 0.15 cm^2/s at the intersection region and a simultaneous localized increase in the PTR amplitude, indicating strong perturbation of heat flow and non-uniform redistribution of absorbed energy. This behavior is consistent with enhanced phonon scattering and localized heat accumulation at regions of concentrated mechanical damage.

From a physical perspective, the PTR phase signal is highly sensitive to variations in thermal diffusivity and local changes in heat propagation length. Therefore, the folding structure observed in the AFM measurements likely introduces localized thermal barriers or scattering regions that slow the propagation of the thermal diffusion wave. Experimentally, this effect appears as a phase delay in the PTR signal near the defect. In contrast, the amplitude response is less sensitive to geometric deformation and primarily reflects variations in local optical absorption and heat generation. The slight amplitude reduction observed near the scratch may, therefore, be associated with changes in local absorption conditions or heat dissipation caused by the surface deformation.

It is important to note that the AFM measurements correspond to a region extending approximately 100 μm around the scratch, which represents a much smaller spatial scale than the PTR mapping. Nevertheless, the presence of localized surface folding and topographic variations provides a plausible physical explanation for the anomalies observed in the PTR phase response at the millimeter scale.

These thermal and photothermal effects are quantitatively supported by XRD results, which show that mechanical damage induces disorders. The slight shift of the (100) diffraction peak from 69.124° to 69.146° reflects the variations in the lattice rather than changes in the intrinsic parameter. Concurrently, the increase in the full width at half maximum (FWHM) from 0.04901 to 0.0601 indicates lattice distortion associated with extrinsic defect generation. Such defects act as effective phonon-scattering centers, leading to reduced thermal diffusivity and anisotropic heat transport. Consequently, the degradation and spatial redistribution of thermal transport measured by PTR and PA techniques can be quantitatively attributed to defect-induced lattice disorder caused by intentional mechanical damage.[6,22]

Boron doping reduces the phonon mean free path (MFP) by introducing additional scattering centers and enhancing carrier–phonon interactions. As the carrier concentration increases, phonon–electron scattering becomes more significant, altering the intrinsic thermal transport of the material. The decrease in phonon MFP directly lowers the lattice thermal conductivity, since heat conduction in crystalline silicon is mainly governed by phonon propagation.

At high doping levels, impurity scattering and carrier-induced scattering further shorten the effective phonon transport length, reducing thermal diffusivity and limiting heat dissipation efficiency. Additionally, defect-related interactions, such as phonon–defect and electron–defect scattering, reduce carrier lifetime and increase momentum relaxation. Although electronic heat transport plays a secondary role in silicon compared to lattice conduction, changes in carrier mobility and lifetime can still affect overall transport behavior, especially in highly doped samples. As a result, lattice deformation and impurity incorporation modify both phononic and electronic contributions to thermal transport, primarily by reducing the phonon mean free path.

E. Electrical properties

Electrical properties were evaluated by Hall effect measurements using the van der Pauw configuration. Table III presents the resistivity values obtained for silicon-01. The silicon-01 sample was analyzed twice to assess resistivity variability. Despite originating from the same material, a marked dispersion is observed in the resistivity of the second specimen, indicating a high degree of variance. This behavior has previously been attributed to variations in electron mobility caused by structural damage that alters the electron mean free path (Parker *et al.*, 2022).[24]

The Hall effect is sensitive to spatial variations in carrier distribution, leading to measurable changes in the carrier concentration. The Hall coefficient (R_H) is expressed as

$$R_H = \frac{\rho e \tau}{m_e^*}, \tag{7}$$

where (ρ) is the electrical resistivity and e. is the elementary charge. Electrical resistivity depends on carrier mobility (μ) and

TABLE III. Electrical parameters on the van der Pauw method and resistivity measurements.

Sample	Current (mA)	Resistivity measured (Ω cm)	Carrier mobility $\left(\frac{\text{cm}}{\text{V s}}\right)$
Silicon-01 (1)	20	0.00511	201. 2
Silicon-01 (2)	20	0.00483	203. 6

effective lifetime (τ), which are governed by scattering mechanisms and the effective mass (m_e^*). The parameter (τ) represents the effective carrier lifetime and is strongly influenced by extrinsic defect.

Carrier mobility was determined by

$$\mu_H = \frac{|R_{Havg}|}{\rho_{avg}}, \tag{8}$$

where the value of R_{Havg} was obtained by averaging R_{HN} and R_{HP}, and the value of the average conductivity (ρ_{avg}) was determined by taking the inverse of the average of the experimentally measured resistivities reported in Table III as the carrier mobility.

The same van der Pauw methodology was applied to the silicon-02 and silicon-03 samples; however, no reliable or reproducible resistivity values were obtained. According to ASTM 76-86, the van der Pauw method requires electrically uniform and continuous samples without significant discontinuities and with dimensions compatible with the electron mean free path. Introducing mechanical scratches violates these conditions by creating localized defects and strong anisotropy in the transport properties. As shown by the evolution of the thermal diffusivity and photothermal distributions (Figs. 6–8), intentional scratches produce localized regions of reduced thermal transport, indicating the disruption of carrier and phonon pathways.

These electrical and thermal transport changes are directly supported by the x-ray diffraction analysis. The slight shift of the (100) diffraction peak from 69.124° in the undamaged wafer to 69.146° after mechanical damage indicates the variations in interplanar spacing, while the increase in the FWHM from 0.04901 to 0.0601 reflects lattice distortion associated with defect generation. Lattice defects act as effective scattering centers for charge carriers, reducing and spatially varying the electron mean free path. Thus, the presence of mechanically induced lattice disorder accounts for both the degradation and heterogeneity of thermal transport and the inability to obtain consistent electrical resistivity values in damaged silicon samples using the van der Pauw method.

As can be inferred, an increased defect density enhances carrier scattering, thereby shortening the carrier lifetime.[6] Consequently, variations in lifetime may lead to apparent changes in the calculated carrier concentration, even if the intrinsic carrier density does not increase; instead, the transport parameters are modified by defect-induced scattering mechanisms.

Similarly, phonon modes are scattered at defect sites and interfaces within the silicon bulk, directly affecting the heat diffusion process. Both charge carriers and phonons are therefore highly sensitive to the structural condition of the material. Changes in the lattice order and increased defect density modify the scattering rates, thereby influencing thermal diffusivity and thermal conductivity through reductions in the phonon mean free path. Thus, mechanically induced lattice disorder accounts for both the degradation and spatial variability of thermal transport, as well as the inability to obtain consistent electrical resistivity values in damaged silicon samples using the van der Pauw method.

IV. CONCLUSIONS

Intrinsic defects in the silicon wafer are primarily related to the spatial distribution of dopants across the wafer, in this case boron, and these defects affect phonon-scattering processes, causing variations in thermal and electrical properties at different locations. Atomic force microscopy (AFM) shows that the intentionally introduced mechanical defect is not spatially uniform due to the lack of controlled force during scratching. The resulting surface morphology displays localized buckling and profile distortion, which promotes additional scattering of both charge carriers and phonons by altering their mean free paths.

X-ray diffraction analysis confirms that mechanical damage causes lattice distortions, as indicated by a slight shift of the Si (100) diffraction peak and an increase in the full width at half maximum (FWHM). Structural refinement shows increase lattice strain after scratching; however, no new crystalline phases form. Instead, the diffraction pattern reveals weak reflections associated with dopant-related phases that become detectable after lattice distortion. These features do not indicate a structural transformation but rather reflect defect-induced lattice perturbations that affect phonon transport.

Photothermal radiometry (PTR) and photoacoustic (PA) results show that both intrinsic and extrinsic defects significantly influence the thermal response of the material. Structural strain and lattice disorder modify optical absorption and heat transport, leading to reduced thermal diffusivity due to enhanced phonon scattering and altered carrier–phonon interactions. Electrical measurements using the van der Pauw method further indicate that cutting-induced defects lower the measured resistivity, while intentionally scratched samples become electrically nonhomogeneous and cannot be reliably characterized.

ACKNOWLEDGMENTS

Porfirio E. Martinez-Munoz and Jose D. Yela-Portilla thank SECIHTI for supporting their Ph.D. studies. This work was supported by the Laboratorio Nacional de Caracterización de Materiales at CFATA-UNAM-Mexico. The authors thank Ph.D. Beatriz Millan-Malo, for the x-ray diffraction pattern measurement. The authors thank Ph.D. Andres de Luna Bugallo and Sofia Peña for the AFM measurements.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Porfirio Esau Martinez-Munoz: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jose D. Yela-Portilla**: Conceptualization (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jonathan**

Y. Lopez-Martinez: Formal analysis (equal); Investigation (equal); Validation (equal); Writing – original draft (equal). **Mario E. Rodríguez-García**: Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1] M. A. Green, *Third Generation Photovoltaics* (Springer, Berlin, 2006).
[2] H. Honarvar, L. Yang, and M. I. Hussein, "Thermal transport size effects in silicon membranes featuring nanopillars as local resonators," Appl. Phys. Lett. **108**, 263101 (2016).
[3] W. Li, F. Zha, B. Fu, Y. Li, J. Duan, and Z. Zhou, "A study on the surface quality and damage properties of single-crystal silicon using different post-treatment processes," Micromachines **15**(1), 145 (2024).
[4] J. Cattin, J. Haschke, C. Ballif, and M. Boccard, "Influence of local surface defects on the minority-carrier lifetime of passivating-contact solar cells," Appl. Phys. Lett. **116**, 113901 (2020).
[5] B. Sopori, S. Devayajanam, and P. Basnyat, "Surface characteristics and damage distributions of diamond wire sawn wafers for silicon solar cells," AIMS Mater. Sci. **3**(2), 669–685 (2016).
[6] H. D. Martinez-Hernandez, P. E. Martinez-Munoz, C. F. Ramirez-Gutierrez, E. U. Martinez-Ascencio, B. M. Millan-Malo, and M. E. Rodriguez-Garcia, "Effect of intrinsic and extrinsic defects on the structural, thermal, and electrical properties in p-type CZ-Si wafers with different carrier concentrations," Int. J. Thermophys. **43**(12), 181 (2022).
[7] T. Lü and Z. Rongzhu, "Effects of intrinsic point defects on response characteristics of silicon," Acta Opt. Sin. **43**(21), 2116002 (2023).
[8] C.-C. Yen, A. K. Singh, Y.-M. Chung, H.-Y. Chou, and D.-S. Wuu, "Study of flow pattern defects and oxidation induced stacking faults in Czochralski single-crystal silicon growth," Crystals **13**(2), 336 (2023).
[9] S. Mirabella, D. De Salvador, E. Napolitani, E. Bruno, and F. Priolo, "Mechanisms of boron diffusion in silicon and germanium," J. Appl. Phys. **113**, 031101 (2013).
[10] A. Semerok, V. Detalle, N. Wilkie-Chancellier, S. Serfaty, X. Bai, T. Calligaro, R. Pillay, E. Ravaud, P. Salinson, W. Pacquentin, and V. Etgens, "Photothermal radiometry for non-destructive testing and characterization of cultural heritage objects," SPIE Proc. 13569, 10 (2025).
[11] M. Commandre, C. Fossati, J.-Y. Natoli, and C. Amra, "Photothermal analysis of submicrometric scale defects in laser damage studies," Proc. SPIE **5250**, 158 (2004).
[12] S. Jeon, J. Kim, J. P. Yun, and C. Kim, "Non-destructive photoacoustic imaging of metal surface defects," J. Opt. **18**(11), 114001 (2016).
[13] Y. Lu and R. Lu, "Non-destructive defect detection of apples by spectroscopic and imaging technologies: A review," Trans. ASABE **60**(5), 1765–1790 (2017).
[14] J. Náhlík, I. Kašpárková, and P. Fitl, "Study of quantitative influence of sample defects on measurements of resistivity of thin films using van der Pauw method," Measurement **44**(10), 1968–1979 (2011).
[15] T. Matsumura and Y. Sato, "A theoretical study on van Der Pauw measurement values of inhomogeneous compound semiconductor thin films," J. Mod. Phys. **01**(05), 340–347 (2010).
[16] P. E. Martinez-Munoz, H. D. Garcia-Vazquez, C. F. Ramirez-Gutierrez, and M. E. Rodriguez-Garcia, "Determination of the thermo-electronic and structural properties of polycrystalline metals and single-crystal semiconductors," Measurement **257**, 118947 (2026).
[17] C. F. Ramirez-Gutierrez, H. D. Martinez-Hernandez, I. A. Lujan-Cabrera, and M. E. Rodriguez-García, "Design, fabrication, and optical characterization of one-dimensional photonic crystals based on porous silicon assisted by in-situ photoacoustics," Sci. Rep. **9**(1), 14732 (2019).
[18] A. C. Bento, N. Cella, S. M. Lima, L. A. O. Nunes, L. H. C. Andrade, J. R. Silva, V. S. Zanuto, N. G. C. Astrath, T. Catunda, A. N. Medina, J. H. Rohling, R. F. Muniz, J. W. Berrar, L. C. Malacarne, W. R. Weinand, F. Sato, M. P. Belancon, G. J. Schiavon, J. Shen, L. C. M. Miranda, H. Vargas, and M. L. Baesso, "Photoacoustic and photothermal and the photovoltaic efficiency of solar cells: A tutorial," J. Appl. Phys. **131**, 141101 (2022).
[19] P. E. Martinez-Munoz, H. D. Martinez-Hernandez, A. Lara-Guevara, B. Millan-Malo, I. Rojas-Rodriguez, and M. E. Rodriguez-Garcia, "Effect of tempering on the metallurgical, structural, and thermal properties of AISI/SAE 1030 steel," Mater. Today Commun. **31**, 103535 (2022).
[20] A. Rosencwaig and A. Gersho, "Theory of the photoacoustic effect with solids," J. Appl. Phys. **47**(1), 64–69 (1976).
[21] M. E. Rodriguez, A. Mandelis, G. Pan, L. Nicolaides, J. A. Garcia, and Y. Riopel, "Computational aspects of laser radiometric multiparameter fit for carrier transport property measurements in Si wafers," J. Electrochem. Soc. **147** (2), 687 (2000).
[22] M. E. Rodrı´guez, J. A. Garcia, and A. Mandelis, "Infrared photothermal radiometry of deep subsurface defects in semiconductor materials," Rev. Sci. Instrum. **74**(1), 839–841 (2003).
[23] P. E. Martinez-Munoz, H. D. Martinez-Hernandez, C. F. Rojas-Beltran, J. L. Perez-Ospina, and M. E. Rodriguez-Garcia, "Development of a differential photoacoustic system for the determination of the effective water diffusion and water vapor permeability coefficients in thin films," J. Appl. Phys. **132**, 115111 (2022).
[24] W. J. Parker, R. J. Jenkins, C. P. Butler, and G. L. Abbott, "Flash method of determining thermal diffusivity, heat capacity, and thermal conductivity," J. Appl. Phys. **32**(9), 1679–1684 (1961).